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Semiconductor device and manufacturing methods thereof

Abstract

Some implementations described herein provide techniques and apparatuses for forming insulator layers in or on a semiconductor substrate prior to forming epitaxial layers within source/drain regions of a fin field-effect transistor. The epitaxial layers may be formed over the insulator layers to reduce electron tunneling between the source/drain regions of the fin field-effect transistor. In this way, a likelihood of leakage into the semiconductor substrate and/or between the source/drain regions of the fin field-effect transistor is reduced.

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Background/Summary

BACKGROUND

(1) Fin-based transistors, such as fin field effect transistors (finFETs) and nanostructure transistors (e.g., nanowire transistors, nanosheet transistors, gate-all-around (GAA) transistors, multi-bridge channel transistors, nanoribbon transistors), are three-dimensional structures that include a channel region in a fin (or a portion thereof) that extends above a semiconductor substrate as a three-dimensional structure. A gate structure, configured to control a flow of charge carriers within the channel region, wraps around the fin of semiconductor material. As an example, in a finFET, the gate structure wraps around three sides of the fin (and thus the channel region), thereby enabling increased control over the channel region (and therefore switching of the finFET). As another example, in a nanostructure transistor, the gate structure wraps around a plurality of channel regions in a fin structure such that the gate structure surrounds each of the plurality of channel regions.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
- (2) FIG. 1 is a diagram of an example environment in which systems and/or methods described herein may be implemented.
- (3) FIG. 2 is a diagram of an example semiconductor structure described herein.
- (4) FIGS. 3A-3H, 4A-4D, and 5A-5F are diagrams of an example implementations described herein.
- (5) FIGS. 6A-6C are diagrams of example types of devices described herein.
- (6) FIGS. 7A-7C are diagrams of example processing operations for forming an insulator layer described herein.
- (7) FIG. 8 is a diagram of an example elemental composition of portions of a semiconductor device described herein.
- (8) FIG. 9 is a diagram of example components of a device described herein.
- (9) FIG. 10 is a flowchart of an example process associated with forming a semiconductor device described herein.

DETAILED DESCRIPTION

- (10) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
- (11) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
- (12) In some cases, reducing geometric and dimensional properties of a fin field-effect transistor (finFET) may decrease a performance of the finFET. As an example, a likelihood of short channel effects, such as drain-induced barrier lowering in a finFET, may increase as finFET technology processing nodes decrease. Additionally or alternatively, a likelihood of electron tunneling and leakage in a finFET may increase as a gate length of the finFET decreases.
- (13) Such leakage may occur within the finFET between source/drain regions of the finFET and/or from the source-drain regions into an underlying semiconductor substrate. The leakage may increase power consumption of a semiconductor device including the finFET and also decrease operational performance of a gate structure between the source/drain regions.
- (14) Some implementations described herein provide techniques and apparatuses for forming

insulator layers in or on a semiconductor substrate prior to forming epitaxial layers within source/drain regions of a finFET. The epitaxial layers may be formed over the insulator layers to reduce electron tunneling between the source/drain regions of the finFET.

(15) In this way, a likelihood of leakage into the semiconductor substrate and/or between the source/drain regions of the finFET is reduced. As a result, performance characteristics relating to power consumption of a semiconductor device including the finFET and operation of a gate structure between the source/drain regions may be increased.

(16) FIG. 1 is a diagram of an example environment **100** in which systems and/or methods described herein may be implemented. As shown in FIG. 1A, environment **100** may include a plurality of semiconductor processing tools **102-114** and a wafer/die transport tool **116**. The plurality of semiconductor processing tools **102-114** may include a deposition tool **102**, an exposure tool **104**, a developer tool **106**, an etch tool **108**, a planarization tool **110**, a plating tool **112**, an ion implantation tool **114**, and/or another type of semiconductor processing tool. The tools included in example environment **100** may be included in a semiconductor clean room, a semiconductor foundry, a semiconductor processing facility, and/or manufacturing facility, among other examples.

(17) The deposition tool **102** is a semiconductor processing tool that includes a semiconductor processing chamber and one or more devices capable of depositing various types of materials onto a substrate. In some implementations, the deposition tool **102** includes a spin coating tool that is capable of depositing a photoresist layer on a substrate such as a wafer. In some implementations, the deposition tool **102** includes a chemical vapor deposition (CVD) tool such as a plasma-enhanced CVD (PECVD) tool, a high-density plasma CVD (HDP-CVD) tool, a sub-atmospheric CVD (SACVD) tool, a low-pressure CVD (LPCVD) tool, an atomic layer deposition (ALD) tool, a plasma-enhanced atomic layer deposition (PEALD) tool, or another type of CVD tool. In some implementations, the deposition tool **102** includes a physical vapor deposition (PVD) tool, such as a sputtering tool or another type of PVD tool. In some implementations, the deposition tool **102** includes an epitaxial tool that is configured to form layers and/or regions of a device by epitaxial growth. In some implementations, the example environment **100** includes a plurality of types of deposition tools **102**.

(18) The exposure tool **104** is a semiconductor processing tool that is capable of exposing a photoresist layer to a radiation source, such as an ultraviolet light (UV) source (e.g., a deep UV light source, an extreme UV light (EUV) source, and/or the like), an x-ray source, an electron beam (e-beam) source, and/or the like. The exposure tool **104** may expose a photoresist layer to the radiation source to transfer a pattern from a photomask to the photoresist layer. The pattern may include one or more semiconductor device layer patterns for forming one or more semiconductor devices, may include a pattern for forming one or more structures of a semiconductor device, may include a pattern for etching various portions of a semiconductor device, and/or the like. In some implementations, the exposure tool **104** includes a scanner, a stepper, or a similar type of exposure tool.

(19) The developer tool **106** is a semiconductor processing tool that is capable of developing a photoresist layer that has been exposed to a radiation source to develop a pattern transferred to the photoresist layer from the exposure tool **104**. In some implementations, the developer tool **106** develops a pattern by removing unexposed portions of a photoresist layer. In some implementations, the developer tool **106** develops a pattern by removing exposed portions of a photoresist layer. In some implementations, the developer tool **106** develops a pattern by dissolving exposed or unexposed portions of a photoresist layer through the use of a chemical developer.

(20) The etch tool **108** is a semiconductor processing tool that is capable of etching various types of materials of a substrate, wafer, or semiconductor device. For example, the etch tool **108** may include a wet etch tool, a dry etch tool, and/or the like. In some implementations, the etch tool **108** includes a chamber that is filled with an etchant, and the substrate is placed in the chamber for a

particular time period to remove particular amounts of one or more portions of the substrate. In some implementations, the etch tool **108** etches one or more portions of the substrate using a plasma etch or a plasma-assisted etch, which may involve using an ionized gas to isotropically or directionally etch the one or more portions.

(21) The planarization tool **110** is a semiconductor processing tool that is capable of polishing or planarizing various layers of a wafer or semiconductor device. For example, a planarization tool **110** may include a chemical mechanical planarization (CMP) tool and/or another type of planarization tool that polishes or planarizes a layer or surface of deposited or plated material. The planarization tool **110** may polish or planarize a surface of a semiconductor device with a combination of chemical and mechanical forces (e.g., chemical etching and free abrasive polishing). The planarization tool **110** may utilize an abrasive and corrosive chemical slurry in conjunction with a polishing pad and retaining ring (e.g., typically of a greater diameter than the semiconductor device). The polishing pad and the semiconductor device may be pressed together by a dynamic polishing head and held in place by the retaining ring. The dynamic polishing head may rotate with different axes of rotation to remove material and even out any irregular topography of the semiconductor device, making the semiconductor device flat or planar.

(22) The plating tool **112** is a semiconductor processing tool that is capable of plating a substrate (e.g., a wafer, a semiconductor device, and/or the like) or a portion thereof with one or more metals. For example, the plating tool **112** may include a copper electroplating device, an aluminum electroplating device, a nickel electroplating device, a tin electroplating device, a compound material or alloy (e.g., tin-silver, tin-lead, and/or the like) electroplating device, and/or an electroplating device for one or more other types of conductive materials, metals, and/or similar types of materials.

(23) An ion implantation tool **114** is a semiconductor processing tool that is capable of implanting ions into a substrate such as a semiconductor wafer. For example, the ion implantation tool **114** generates ions in an arc chamber from a source material such as a gas or a solid. The source material is provided into the arc chamber, and an arc voltage is discharged between a cathode and an electrode to produce a plasma containing ions of the source material. One or more extraction electrodes are used to extract the ions from the plasma in the arc chamber and accelerate the ions to form an ion beam. The ion beam may be directed toward the substrate such that the ions are implanted below the surface of the substrate to dope the substrate.

(24) Wafer/die transport tool **116** includes a mobile robot, a robot arm, a tram or rail car, an overhead hoist transport (OHT) system, an automated materially handling system (AMHS), and/or another type of device that is configured to transport substrates and/or semiconductor devices between semiconductor processing tools **102-114**, that is configured to transport substrates and/or semiconductor devices between processing chambers of the same semiconductor processing tool, and/or that is configured to transport substrates and/or semiconductor devices to and from other locations such as a wafer rack, a storage room, and/or the like. In some implementations, wafer/die transport tool **116** includes a programmed device that is configured to travel a particular path and/or may operate semi-autonomously or autonomously. In some implementations, the semiconductor processing environment **100** includes a plurality of wafer/die transport tools **116**.

(25) The wafer/die transport tool **116** may be included in a cluster tool or another type of tool that includes a plurality of processing chambers, and may be configured to transport substrates and/or semiconductor devices between the plurality of processing chambers, to transport substrates and/or semiconductor devices between a processing chamber and a buffer area, to transport substrates and/or semiconductor devices between a processing chamber and an interface tool such as an equipment front end module (EFEM), and/or to transport substrates and/or semiconductor devices between a processing chamber and a transport carrier (e.g., a front opening unified pod (FOUP)), among other examples. In some implementations, a wafer/die transport tool **116** may be included in a multi-chamber (or cluster) deposition tool **102**, which may include a pre-clean processing

chamber (e.g., for cleaning or removing oxides, oxidation, and/or other types of contamination or byproducts from a substrate and/or semiconductor device) and a plurality of types of deposition processing chambers (e.g., processing chambers for depositing different types of materials, processing chambers for performing different types of deposition operations). In these implementations, the wafer/die transport tool **116** is configured to transport substrates and/or semiconductor devices between the processing chambers of the deposition tool **102** without breaking or removing a vacuum (or an at least partial vacuum) between the processing chambers and/or between processing operations in the deposition tool **102**, as described herein.

(26) The number and arrangement of devices shown in FIG. **1** are provided as one or more examples. In practice, there may be additional devices, fewer devices, different devices, or differently arranged devices than those shown in FIG. **1**. Furthermore, two or more devices shown in FIG. **1** may be implemented within a single device, or a single device shown in FIG. **1** may be implemented as multiple, distributed devices. Additionally, or alternatively, a set of devices (e.g., one or more devices) of environment **100** may perform one or more functions described as being performed by another set of devices of environment **100**.

(27) FIG. **2** is a diagram of an example semiconductor device **200** structure described herein. The semiconductor device **200** includes a device region **202**. Example types of the semiconductor device **200** include a static random access memory (SRAM) type of semiconductor device, an input/output (IO) type of semiconductor type of device, or a ring oscillator (RO) type of semiconductor device, among other examples.

(28) The device region **202** may include one or more transistors or other devices. The transistors may include fin-based transistors, such as fin field effect transistors (finFETs), nanostructure transistors, and/or other types of transistors. In some implementations, the device region **202** includes a p-type metal oxide semiconductor (PMOS) region, an n-type metal oxide semiconductor (NMOS) region, a complementary metal oxide semiconductor (CMOS) region, and/or another type of device region.

(29) The device region **202** includes a semiconductor substrate **204**. The semiconductor substrate **204** includes a silicon (Si) substrate, a substrate formed of a material including silicon, a III-V compound semiconductor material substrate such as gallium arsenide (GaAs), a silicon on insulator (SOI) substrate, a germanium (Ge) substrate, a silicon germanium (SiGe) substrate, or another type of semiconductor substrate.

(30) Fin structures **206** are included above (and/or extend above) the semiconductor substrate **204** for the device region **202**. A fin structure **206** may provide an active region where one or more devices (e.g., fin-based transistors) are formed, and may therefore be referred to as active fin structures. In some implementations, the fin structures **206** include silicon (Si) materials or another elementary semiconductor material such as germanium (Ge). In some implementations, the fin structures **206** include an alloy semiconductor material such as silicon germanium (SiGe), gallium arsenide phosphide (GaAsP), aluminum indium arsenide (AlInAs), aluminum gallium arsenide (AlGaAs), gallium indium arsenide (GaInAs), gallium indium phosphide (GaInP), gallium indium arsenide phosphide (GaInAsP), or a combination thereof. In some implementations, the fin structures **206** are doped using n-type and/or p-type dopants.

(31) The fin structures **206** are fabricated by suitable semiconductor process techniques, such as masking, photolithography, and/or etch processes, among other examples. As an example, the fin structures **206** may be formed by etching a portion of the semiconductor substrate **204** away to form recesses in the semiconductor substrate **204**. The recesses may then be filled with isolating material that is recessed or etched back.

(32) In some implementations, hybrid fin structures **208** are included between the fin structures **206** in an alternating arrangement. The hybrid fin structures **208** may also be referred to as h-fins, dummy fins, and/or non-active fins, among other examples. A hybrid fin structure **208** includes a fin structure that extends in the first direction approximately parallel to the fin structures **206**. In

some implementations, a hybrid fin structure **208** is included between two fin structures **206** and extends approximately the same length as the two fin structures **206**.

(33) A hybrid fin structure **208** is configured to provide electrical isolation between one or more structures and/or components included in the device region **202**. In some implementations, a hybrid fin structure **208** is configured to provide electrical isolation between two or more fin structures **206** (e.g., two or more active fin structures). In some implementations, a hybrid fin structure **208** is configured to provide electrical isolation between two or more source/drain regions in the device region **202**. In some implementations, a hybrid fin structure **208** is configured to provide electrical isolation between two or more gates structures or two or more portions of a gate structure. In some implementations, a hybrid fin structure **208** is configured to provide electrical isolation between a source/drain region and a gate structure.

(34) A hybrid fin structure **208** described herein includes a plurality of types of dielectric materials. A hybrid fin structure **208** may include a combination of one or more low dielectric constant (low-k) dielectric materials (e.g., a silicon oxide (SiO_{2-x}) and/or a silicon nitride (Si_xN_y), among other examples) and a one or more high dielectric constant (high-k) dielectric materials (e.g., a hafnium oxide (HfO_x) and/or other high-k dielectric material).

(35) Shallow trench isolation (STI) regions **210** are included above the semiconductor substrate **204** and between the fin structures **206** and the hybrid fin structures **208**. The STI regions **210** may be formed by etching back an insulating layer formed on the fin structures **206** and on the semiconductor substrate **204**. However, other fabrication techniques for the STI regions **210** may be used. The STI regions **210** may electrically isolate adjacent active areas in the fin structures **206** and/or adjacent portions of the fin structures **206** and the hybrid fin structures **208**. The STI regions **210** may include a dielectric material such as a silicon oxide (SiO_{2-x}), a silicon nitride (Si_xN_y), a silicon oxynitride (SiON), fluoride-doped silicate glass (FSG), a low-k dielectric material, and/or other suitable insulating material. The STI regions **210** may include a multi-layer structure, for example, having one or more liner layers.

(36) A dummy gate structure **212** (or a plurality of dummy gate structures **212**) is included in the device region **202** over the fin structures **206** (e.g., approximately perpendicular to the fin structures **206**). The dummy gate structure **212** engages the fin structures **206** on three or more sides of the fin structures **206**. Moreover, the dummy gate structure **212** may be included over the hybrid fin structures **208** and may engage the hybrid fin structures **208** on three or more sides of the hybrid fin structures **208**. In the example depicted in FIG. 2, the dummy gate structure **212** includes a gate dielectric layer **214**, a gate electrode layer **216**, and a hard mask layer **218**. In some implementations, the dummy gate structure **212** further includes a capping layer, one or more spacer layers, and/or another suitable layer. The various layers of the dummy gate structure **212** may be formed by suitable deposition techniques and patterned by suitable photolithography and etching techniques.

(37) The term “dummy,” as described herein, refers to a sacrificial structure which will be removed in a later stage and will be replaced with another structure, such as a high-k dielectric and metal gate structure in a replacement gate process. The replacement gate process refers to manufacturing a gate structure at a later stage of the overall gate manufacturing process. Accordingly, the configuration of the device region **202** illustrated in FIG. 2 may include an intermediate configuration, and additional semiconductor processing operations may be performed for the device region **202** to further process the device region **202**.

(38) The gate dielectric layer **214** may include a dielectric oxide layer. The dielectric oxide layer may be formed by chemical oxidation, thermal oxidation, ALD, CVD, and/or other suitable methods. The gate electrode layer **216** may include a poly-silicon material or another suitable material. The gate electrode layer **216** may be formed by suitable deposition processes such as LPCVD or PECVD, among other examples. The hard mask layer **218** may include any material suitable to pattern the gate electrode layer **216** with particular features/dimensions on the

semiconductor substrate **204**.

(39) In some implementations, the various layers of the dummy gate structure **212** are first deposited as blanket layers. Then, the blanket layers are patterned through a process including photolithography and etching processes, removing portions of the blanket layers and keeping the remaining portions over the STI regions **210**, over the fin structures **206**, and/or over the hybrid fin structures **208** to form the dummy gate structure **212**.

(40) Source/drain areas **220** are disposed in opposing regions of the fin structures **206** (e.g., in opposing regions of the active fin structures) with respect to the dummy gate structure **212**. The source/drain areas **220** include areas in the device region **202** in which source/drain regions are to be formed in one or more epitaxial growth operations, and therefore the source/drain regions may be referred to as epitaxial regions. The source/drain regions in the device region **202** include silicon (Si) with one or more dopants, such as a p-type material (e.g., boron (B) or germanium (Ge), among other examples), an n-type material (e.g., phosphorous (P) or arsenic (As), among other examples), and/or another type of dopant. Accordingly, the device region **202** may include PMOS transistors that include p-type source/drain regions, NMOS transistors that include n-type source/drain regions, and/or other types of transistors.

(41) Some source/drain regions may be shared between various transistors in the device region **202**. In some implementations, various ones of the source/drain regions may be connected or coupled together such that fin-based transistors in the device region **202** are implemented as two functional transistors. For example, if neighboring (e.g., as opposed to opposing) source/drain regions are electrically connected, such as through coalescing the source/drain regions by epitaxial growth (e.g., neighboring source/drain regions, as opposed to on opposing sides of the dummy gate structure **212**, being coalesced), two functional transistors may be implemented. Coalesced neighboring source/drain regions are referred to herein as merged epitaxial regions and merged source/drain regions. Other configurations in other examples may implement other numbers of functional transistors.

(42) FIG. 2 further illustrates reference cross-sections that are used in later figures, including FIGS. 3A-7C. FIGS. 3A-7C are schematic cross-sectional views of various portions of the device region **202** and/or another device region of the semiconductor device **200** illustrated in FIG. 2, and correspond to various processing stages of forming fin-based transistors in the device region **202** and/or another device region of the semiconductor device **200**. Cross-section A-A is in a plane along a channel in a fin structure **206** between opposing source/drain areas **220**. Cross-section B-B is in a plane perpendicular to cross-section A-A, and is across a plurality of fin structures **206** and across a plurality of hybrid fin structures **208** in a plurality of source/drain areas **220**. Subsequent figures refer to these reference cross-sections for clarity. In some figures, some reference numbers of components or features illustrated therein may be omitted to avoid obscuring other components or features for ease of depicting the figures.

(43) As indicated above, FIG. 2 is provided as an example. Other examples may differ from what is described with regard to FIG. 2.

(44) FIGS. 3A-3H are diagrams of an example implementation **300** described herein. The example implementation **300** includes an example of forming the fin structures **206** and forming the hybrid fin structures **208** for transistors in the device region **202** (and/or another device region) of the semiconductor device **200**. FIGS. 3A-3H are illustrated from the perspective of the cross-sectional plane B-B in FIG. 2 for the device region **202**.

(45) Turning to FIG. 3A, the example implementation **300** includes semiconductor processing operations relating to the semiconductor substrate **204** in and/or on which transistors are formed in the device region **202**. The deposition tool **102** forms a hard mask layer **302** on the semiconductor substrate **204** using a CVD technique, a PVD technique, a spin-coating technique, and/or another deposition technique described above in connection with FIG. 1. The hard mask layer **302** includes a material having an etch selectivity to particular types of etchants that is different relative to the

material of the semiconductor substrate **204** to facilitate etching of the semiconductor substrate **204**. For example, the hard mask layer **302** may include a silicon nitride (Si.sub.xN.sub.y) or another suitable material, and the semiconductor substrate **204** may include a silicon oxide (SiO.sub.x) or another suitable material.

(46) In the example implementation **300**, an insulator layer **304** is included within the semiconductor substrate **204**. As described in greater detail in connection with FIGS. 7A-7C, different techniques using one or more combinations of the semiconductor processing tools **102-114** may form the insulator layer **304**. The insulator layer **304** may include a dielectric material such as a silicon oxide (SiO.sub.x) or another suitable material. The combination of the semiconductor substrate **204** and the insulator layer **304**, in some implementations, is referred to as a silicon-on-insulator (SOI) substrate.

(47) As shown in FIG. 3B, the fin structures **206** are formed in the semiconductor substrate **204** in the device region **202**. In some implementations, a pattern in a photoresist layer is used to form the fin structures **206**. In these implementations, the deposition tool **102** forms the photoresist layer on the hard mask layer **302**. The exposure tool **104** exposes the photoresist layer to a radiation source to pattern the photoresist layer. The developer tool **106** develops and removes portions of the photoresist layer to expose the pattern. The etch tool **108** etches into the hard mask layer **302** to form a pattern in the hard mask layer **302**. The semiconductor substrate **204** is then etched based on the pattern in the hard mask layer **302** to form the fin structures **206** in a portion of the semiconductor substrate **204**. In some implementations, the etch operation includes a plasma etch technique, a wet chemical etch technique, and/or another type of etch technique. In some implementations, a photoresist removal tool removes the remaining portions of the photoresist layer (e.g., using a chemical stripper, plasma ashing, and/or another technique). Each fin structure **206** includes a portion of the insulator layer **304**.

(48) As shown in FIG. 3C, a dielectric layer **306** is formed over and/or on the semiconductor substrate **204** and over and/or on the fin structures **206** including the portions of the insulator layer **304**. The deposition tool **102** deposits the dielectric layer **306** using a CVD technique, a PVD technique, an ALD technique, or another type of deposition technique described above in connection with FIG. 1. In some implementations, the dielectric layer **306** is conformally deposited on the fin structures **206** such that the dielectric layer **306** is formed on the tops and sidewalls of the fin structures **206**.

(49) As shown in FIG. 3D, a low-k dielectric material layer **308** is formed in recesses in the dielectric layer **306** between the fin structures **206**. The deposition tool **102** deposits the low-k dielectric material layer **308** using a CVD technique, a PVD technique, an ALD technique, or another type of deposition technique described above in connection with FIG. 1. In some implementations, the low-k dielectric material layer **308** is formed in a dielectric merge operation in which separate portions of the low-k dielectric material layer **308** are merged together.

(50) As shown in FIG. 3E, the low-k dielectric material layer **308** is etched back such that a height of a top surface of the low-k dielectric material layer **308** is lesser relative to a height of a top surface of the dielectric layer **306**. The etch tool **108** etches the low-k dielectric material layer **308** using a plasma etch technique, a wet chemical etch technique, and/or another type of etch technique. In some implementations, the low-k dielectric material layer **308** is etched back such that a height of a top surface of the low-k dielectric material layer **308** is greater relative to a height of a top surface of the hard mask layer **302**, and is greater relative to a height of a top surface of the fin structures **206**. In some implementations, the low-k dielectric material layer **308** is etched back such that a height of a top surface of the low-k dielectric material layer **308** is lesser relative to the height of a top surface of the hard mask layer **302**, and is greater relative to the height of a top surface of the fin structures **206**. In some implementations, the low-k dielectric material layer **308** is etched back such that a height of a top surface of the low-k dielectric material layer **308** is lesser relative to the height of a top surface of the hard mask layer **302**, and is lesser relative to the height

of a top surface of the fin structures **206**.

(51) As shown in FIG. 3F, a high-k dielectric material layer **310** is formed in recesses in the dielectric layer **306** between the fin structures **206** including the portions of the insulator layer **304**. Moreover, the high-k dielectric material layer **310** is formed on the low-k dielectric material layer **308**. The deposition tool **102** deposits the high-k dielectric material layer **310** using a CVD technique, a PVD technique, an ALD technique, or another type of deposition technique described above in connection with FIG. 1. In some implementations, the high-k dielectric material layer **310** is formed in a dielectric merge operation in which separate portions of the high-k dielectric material layer **310** are merged together.

(52) As shown in FIG. 3G, the high-k dielectric material layer **310** is etched to form the hybrid fin structures **208**. The hybrid fin structures **208** each include a portion of the low-k dielectric material layer **308** and a portion of the high-k dielectric material layer **310**. The high-k dielectric material layer **310** is etched such that a height of a top surface of the high-k dielectric material layer **310** is lesser relative to a height of a top surface of the dielectric layer **306**. The etch tool **108** etches the low-k dielectric material layer **308** using a plasma etch technique, a wet chemical etch technique, and/or another type of etch technique.

(53) In some implementations, the high-k dielectric material layer **310** is etched back such that a height of a top surface of the high-k dielectric material layer **310** is greater relative to a height of a top surface of the hard mask layer **302**, and is greater relative to a height of a top surface of the fin structures **206**. In some implementations, the high-k dielectric material layer **310** is etched back such that a height of a top surface of the high-k dielectric material layer **310** is lesser relative to the height of a top surface of the hard mask layer **302**, and is greater relative to the height of a top surface of the fin structures **206**. In some implementations, the high-k dielectric material layer **310** is etched back such that a height of a top surface of the high-k dielectric material layer **310** is lesser relative to the height of a top surface of the hard mask layer **302**, and is lesser relative to the height of a top surface of the fin structures **206**.

(54) As shown in FIG. 3H, the dielectric layer **306** is etched to form the STI regions **210** between adjacent fin structures **206** and/or between adjacent pairs of a fin structure **206** and a hybrid fin structure **208**. In this way, portions of the fin structures **206** are exposed so that the epitaxial regions (or source/drain regions) can be formed on the exposed portion of the fin structures **206** above the STI regions **210**. The dielectric layer **306** is etched such that top surfaces of the STI regions **210** are below the tops of the fin structures **206**. Moreover, the dielectric layer **306** is etched such that top surfaces of the STI regions **210** are above the top surfaces of the portions of the insulator layer **304** in fin structures **206**. In this way, the portions of the insulator layer **304** are below the top surfaces of the STI regions **210**.

(55) In some implementations, a pattern in a photoresist layer is used to etch the dielectric layer **306** to form the STI regions **210**. In these implementations, the deposition tool **102** forms the photoresist layer on the dielectric layer **306** and on the high-k dielectric material layer **310**. The exposure tool **104** exposes the photoresist layer to a radiation source to pattern the photoresist layer. The developer tool **106** develops and removes portions of the photoresist layer to expose the pattern. The etch tool **108** etches the dielectric layer **306** based on the pattern to form the STI regions **210**. In some implementations, the etch operation includes a plasma etch technique, a wet chemical etch technique, and/or another type of etch technique. In some implementations, a photoresist removal tool removes the remaining portions of the photoresist layer (e.g., using a chemical stripper, plasma ashing, and/or another technique). In some implementations, a hard mask layer is used as an alternative technique for forming the STI regions **210** based on a pattern. In some implementations, the dielectric layer **306** is etched based on the etch selectivity between the material of the dielectric layer **306** and the material of the high-k dielectric material layer **310** (e.g., without a hard mask or a photoresist layer on the high-k dielectric material layer **310**).

(56) As indicated above, FIGS. 3A-3H are provided as an example. Other examples may differ

from what is described with regard to FIGS. 3A-3H.

(57) FIGS. 4A-4D are diagrams of an example implementation **400** described herein. The example implementation **400** includes an example of forming dummy gate structures and forming additional layers (e.g., one or more spacer layers of material) over the fin structures **206** including respective portions of the insulator layer **304**. FIGS. 4A-4D are illustrated from one or more perspectives illustrated in FIG. 2, including the perspective of the cross-sectional plane A-A in FIG. 2 for the device region **202** and the perspective of the cross-sectional plane B-B in FIG. 2 for the device region **202**. In some implementations, processing operations described in connection with the example implementation **400** are performed after the fin formation process described in connection with FIGS. 3A-3H and performed by one or more of the semiconductor processing tools **102-114**.

(58) As shown in FIG. 4A, the dummy gate structures **212** are formed in the device region **202**. The dummy gate structures **212** are formed and included over the fin structures **206**, and around the sides of the fin structures **206** such that the dummy gate structures **212** surround the fin structure **206** on at least three sides of the fin structure **206**. Similarly, the dummy gate structures **212** are formed and included over the hybrid fin structures **208**, and around the sides of the hybrid fin structures **208** such that the dummy gate structures **212** surround the hybrid fin structures **208** on at least three sides of the hybrid fin structures **208**.

(59) The dummy gate structures **212** are formed as placeholders for the actual gate structures (e.g., replacement high-k gate structures or metal gate structures) that are to be formed for the transistors included in the device region **202**. The dummy gate structures **212** may be formed as part of a replacement gate process, which enables other layers and/or structures to be formed prior to formation of the replacement gate structures.

(60) The dummy gate structures **212** include gate dielectric layers **214**, gate electrode layers **216**, and hard mask layers **218**. The gate dielectric layers **214** may each include dielectric oxide layers. As an example, the gate dielectric layers **214** may each be formed (e.g., by the deposition tool **102**) by chemical oxidation, thermal oxidation, ALD, CVD, and/or one or more other suitable methods. The gate electrode layers **216** may each include a poly-silicon layer or one or more other suitable layers. For example, the gate electrode layers **216** may be formed (e.g., by the deposition tool **102**) by suitable deposition processes such as LPCVD or PECVD, among other examples. The hard mask layers **218** may each include any material suitable to pattern the gate electrode layers **216** with particular dimensions and/or attributes. Examples include silicon nitride, silicon oxynitride, silicon carbon nitride, or a combination thereof, among other examples. The hard mask layers **218** may be deposited (e.g., by the deposition tool **102**) by CVD, PVD, ALD, or another deposition technique.

(61) As further shown in FIG. 4A, seal spacer layers **402** are included on the sidewalls of the dummy gate structures **212**. The seal spacer layers **402** may be conformally deposited (e.g., by the deposition tool **102**) and may include a silicon oxycarbide (SiOC), a nitrogen free SiOC, or another suitable material. The seal spacer layers **402** may be formed using an ALD operation in which various types of precursor gasses including silicon (Si) and carbon (C) are sequentially supplied in a plurality of alternating cycles to form the seal spacer layers **402**, among other example deposition techniques.

(62) As further shown in FIG. 4A, bulk spacer layers **404** may be formed on the seal spacer layers **402**. The bulk spacer layers **404** may be formed of similar materials as the seal spacer layers **402**. However, the bulk spacer layers **404** may be formed without plasma surface treatment that is used for the seal spacer layers **402**. Moreover, the bulk spacer layers **404** may be formed to a greater thickness relative to the thickness of the seal spacer layers **402**.

(63) In some implementations, the seal spacer layers **402** and the bulk spacer layers **404** are conformally deposited (e.g., by the deposition tool **102**) on the dummy gate structures **212**, and on the fin structures **206**. The seal spacer layers **402** and the bulk spacer layers **404** are then patterned (e.g., by the deposition tool **102**, the exposure tool **104**, and the developer tool **106**) and etched

(e.g., by the etch tool **108**) to remove the seal spacer layers **402** and the bulk spacer layers **404** from the tops of the dummy gate structures **212**, from the fin structures **206**, and from the hybrid fin structures **208**.

(64) As shown in FIG. **4B** and FIG. **4C**, a spacer layer **406** is formed on the fin structures **206**, on the hybrid fin structures **208**, and on the STI regions **210**. The spacer layer **406** includes a silicon oxycarbide (SiOC), a nitrogen free SiOC, a silicon nitride (Si.sub.xN.sub.y), and/or another suitable material. In some implementations, the spacer layer **406** includes a plurality of spacer layers. In some implementations, the spacer layer **406** and one or more of the seal spacer layers **402** and/or the bulk spacer layers **404** are the same spacer layer(s). The deposition tool **102** deposits the spacer layer **406** using a CVD technique, a PVD technique, an ALD technique, or another type of deposition technique described above in connection with FIG. **1**. As shown in FIGS. **4B** and **4C**, the spacer layer **406** is conformally deposited on the top surfaces and the sidewalls of the fin structures **206**, on the top surfaces and the sidewalls of the hybrid fin structures **208**, and on the top surfaces of the STI regions **210**.

(65) As shown in FIG. **4D**, portions of the spacer layer **406** are removed from the top surfaces of the fin structures **206** and from the top surfaces of the hybrid fin structures **208**. The etch tool **108** may remove the portions of the spacer layer **406**. The etch operation to remove the portions of the spacer layer **406** may be referred to as a fin top opening operation, and may be performed to facilitate etching of the fin structures **206**.

(66) As indicated above, FIGS. **4A-4D** are provided as an example. Other examples may differ from what is described with regard to FIGS. **4A-4D**.

(67) FIGS. **5A-5F** are diagrams of an example implementation **500** described herein. In connection with forming the fin structure **206** in the semiconductor substrate **204** as described in connection with FIGS. **3A-3H** and **4A-4D**, the example implementation **500** includes processing operations to form epitaxial regions that correspond to source/drain regions of transistors included in the device region **202**. FIGS. **5A-5F** are illustrated from the perspective of the cross-sectional plane B-B in FIG. **2** for the device region **202**. The example implementation **500** further includes processing operations to form a metal source/drain contact (MD) structure included in the device region **202**.

(68) FIG. **5A** shows a fin structure **206a**, a fin structure **206b**, and a hybrid fin structure **208** above the semiconductor substrate **204**. The STI region **210** is between the fin structure **206a** and the fin structure **206b**. Below the fin structure **206a** is a portion of the insulator layer **304a** (e.g., the insulator layer **304a** includes a dielectric layer, such as an oxide layer among other examples, and may be included as part of the fin structure **206a**). The portion of the insulator layer **304a**, as shown, may be located at or near a bottom of the fin structure **206a**. Another portion of the insulator layer **304b** (e.g., the insulator layer **304b** includes a dielectric layer, such as an oxide layer among other examples, and may be included as part of the fin structure **206b**). The portion of the insulator layer **304b**, as shown, may be located at or near a bottom of the fin structure **206b**. Further, and as shown in FIG. **5A**, a plurality of spacer layers **406** are on sidewalls of the fin structures **206a** and **206b**.

(69) FIG. **5B** shows forming recesses **502** in the fin structures **206**. As an example, the etch tool **108** of FIG. **1** may selectively etch the fin structures **206** to form the recesses **502**. Selectively etching the fin structures **206** may also include selectively etching the spacer layers **406** to form one or more fin sidewall spacers **504**. The etch tool **108** may perform a strained source/drain (SSD) etch operation to form the recesses **502**. Accordingly, the recesses **502** may be referred to as SSD recesses.

(70) In some implementations, and as shown, the recesses **502** are formed to, or into, portions of the insulator layers **304a** and **304b**. This enables the source/drain regions to be formed over and/or on the insulator layers **304a** and **304b**. In this way, the insulator layers **304a** and **304b** are configured to reduce and/or minimize leakage (e.g., current leakage, electron leakage, electron hole leakage) under the source/drain regions. In some implementations, the recesses **502** are formed to

top surfaces of the insulator layers **304a** and **304b**. In some implementations, the recesses **502** are formed into a portion of the insulator layers **304a** and **304b**. As a result, the top surfaces of insulator layers **304a** and **304b** may be concave and/or rounded. A top surface of the recesses **502** may be formed to a depth **506** that is in a range from approximately 5 nanometers to approximately 30 nanometers below a top surface of the STI region **210**. For a depth **506** that is less than this range, a transistor in the device region **202** may not function electrically, while for a depth **506** that is greater than this range, leakage from the transistor may increase. However, other values and ranges for the depth **506** are within the scope of the present disclosure.

(71) FIG. 5C shows forming epitaxial layers **508** (e.g., the epitaxial layer **508a** and the epitaxial layer **508b**) on the insulator layers **304** and between the fin sidewall spacers **504**. In some implementations, the epitaxial layers **508** are lightly doped. The epitaxial layer **508a** extends into the fin structure **206a** and the epitaxial layer **508b** extends into the fin structure **206b**. A combination of processing operations, including deposition processing operations performed by the deposition tool **102**, etching processing operations performed by the etch tool **108** of FIG. 1, among other examples, may form the epitaxial layers **508**.

(72) For an implementation where the fin structures **206** correspond to NMOS-type fins, forming the epitaxial layers **508** may include depositing and etching a silicon material including an arsenic dopant (SiAs). As an example, a concentration of the arsenic dopant may be in a range from approximately 1×10^{19} atoms per cubic centimeter to approximately 5×10^{20} atoms per cubic centimeter. However, other values and ranges for the concentration of the arsenic dopant in the epitaxial layers **508** of NMOS-type fins are within the scope of the present disclosure.

(73) For an implementation where the fin structures **206** correspond to PMOS-type fins, forming the epitaxial layers **508** may include depositing and etching a silicon-germanium material including a boron dopant (SiGeB). As an example, a concentration of the boron dopant may be in a range from approximately 8×10^{19} atoms per cubic centimeter to approximately 3×10^{20} atoms per cubic centimeter. Additionally, or alternatively, a content of germanium may range from approximately 15% to approximately 18%. However, other values and ranges for the concentration of the boron dopant or the content of germanium in the epitaxial layers **508** of PMOS-type fins are within the scope of the present disclosure.

(74) A thickness of the epitaxial layers **508** (e.g., relative to a top surface of the insulator layers **304**) may be in a range from approximately 3 nanometers to approximately 15 nanometers. However, other values and ranges for the thickness of the epitaxial layers **508** are within the scope of the present disclosure.

(75) FIG. 5D shows forming epitaxial layers **510** (e.g., the epitaxial layer **510a** and the epitaxial layer **510b**) on the epitaxial layers **508** above the fin sidewall spacers **504**. In some implementations, the epitaxial layers **510** are more heavily doped than the epitaxial layers **508**. As shown, epitaxial layers **510** combine with the epitaxial layers **508** to form epitaxial regions **512** (e.g., epitaxial region **512a** and epitaxial region **512b**). Forming the epitaxial layers **510** may include forming the merged-epitaxial sub-region **514**. The merged-epitaxial sub-region **514** may join the epitaxial region **512a** to the epitaxial region **512b**.

(76) Forming the epitaxial layers **510** may include a combination of processing operations, including deposition operations performed by the deposition tool **102** and etching operations performed by the etch tool **108** of FIG. 1. As an example, and for an implementation where the fin structures **206** correspond to NMOS-type fins, forming the epitaxial layers **510** may include depositing and etching a silicon material including a phosphorous dopant (SiP). A concentration of the phosphorous dopant may be in a range from approximately 1×10^{20} atoms per cubic centimeter to approximately 5×10^{21} atoms per cubic centimeter. However, other values and ranges for the concentration for the phosphorous dopant in the epitaxial layers **510** of NMOS-type fins are within the scope of the present disclosure.

(77) As another example, and for an implementation where the fin structures **206** correspond to

PMOS-type fins, forming the epitaxial layers **510** may include depositing and etching a silicon-germanium material including a boron dopant (SiGeB). As an example, a concentration of the boron dopant may be in a range from approximately 4×10^{20} atoms per cubic centimeter to approximately 1×10^{21} atoms per cubic centimeter. Additionally, or alternatively, a content of germanium may range from approximately 40% to approximately 60%. However, other values and ranges for the concentration of the boron dopant or the content of germanium in the epitaxial layers **510** of PMOS-type fins are within the scope of the present disclosure.

(78) A thickness of the epitaxial layers **510** (e.g., relative to a top surface of the epitaxial layers **508**) may be in a range from approximately 30 nanometers to approximately 70 nanometers.

However, other values and ranges for the thickness of the epitaxial layers **510** are within the scope of the present disclosure.

(79) FIG. 5E shows an epitaxial layer **516** (e.g., a capping layer) formed over the epitaxial layers **510**. A combination of processing operations, including deposition operations performed by the deposition tool **102** and etching operations performed by the etch tool **108** of FIG. 1, may form the epitaxial layer **516**. As an example, and for an implementation where the fin structures **206** correspond to NMOS-type fins, forming the epitaxial layer **516** may include depositing and etching a silicon material including a phosphorous dopant (SiP). A concentration of the phosphorous dopant may be in a range from approximately 1×10^{20} atoms per cubic centimeter to approximately 2×10^{21} atoms per cubic centimeter. However, other values and ranges for the concentration for the phosphorous dopant in the epitaxial layers **516** of NMOS-type fins are within the scope of the present disclosure.

(80) As another example, and for an implementation where the fin structures **206** correspond to PMOS-type fins, forming the epitaxial layer **516** may include depositing and etching a silicon-germanium material including a boron dopant (SiGeB). As an example, a concentration of the boron dopant may be in a range from approximately 8×10^{19} atoms per cubic centimeter to approximately 8×10^{20} atoms per cubic centimeter. Additionally, or alternatively, a content of germanium may range from approximately 10% to approximately 20%. However, other values and ranges for the concentration of the boron dopant or the content of the germanium in the epitaxial layers **516** of PMOS-type fins are within the scope of the present disclosure.

(81) A thickness of the epitaxial layer **516** may be in a range from approximately 3 nanometers to approximately 10 nanometers. However, other values and ranges for the thickness of the epitaxial layer **516** are within the scope of the present disclosure.

(82) FIG. 5F shows additional layers of materials which may be formed over the fin structures **206**. The additional layers, which include an inner-layer dielectric (ILD) layer **518**, a contact layer **520**, and a conductive layer **522**, may correspond to a metal source/drain contact (MD) structure.

(83) A combination of processing operations, including deposition operations performed by the deposition tool **102** and etching operations performed by the etch tool **108** of FIG. 1, may form the layers **518-522**. The deposition and etching operations may include depositing the ILD layer **518**, etching a recess in the ILD layer **518**, selectively etching tops of the epitaxial regions **512** (including the epitaxial layer **516**), and depositing the contact layer **520**. After deposition of the contact layer **520**, the deposition tool **102** may fill the recess in the ILD layer **518** with the conductive layer **522** using a PVD operation. In some implementations, the planarization tool **110** of FIG. 1 performs a chemical-mechanical polish (CMP) operation to control a height of one or more portions of the MD contact structure (e.g., a height of the ILD layer **518** or a height of the conductive layer **522**, among other examples).

(84) In some implementations, the ILD layer **518** includes an aluminum oxynitride (AlON) material, a silicon nitride (Si.sub.xN.sub.y) material, or a silicon oxide (Si.sub.xO.sub.y) material, among other examples. In some implementations, the contact layer **520** includes a titanium (Ti) material. In some implementations, the conductive layer **522** includes a cobalt (Co) material, a tungsten (W) material, and/or a ruthenium (Ru) material, among other examples. However, other

types of materials for the ILD layer **518**, the contact layer **520**, and the conductive layer **522** are within the scope of the present disclosure.

(85) Processing operations of FIGS. **5A-5F**, in connection with processing operations described in FIGS. **3A-3H** and **4A-4D**, may be performed to form the semiconductor device **200**. The semiconductor device **200** includes the fin structure **206a** (e.g., a first fin structure) above the semiconductor substrate **204** and the fin structure **206b** (e.g., a second fin structure) above the semiconductor substrate **204** and adjacent to the fin structure **206a**. The semiconductor device **200** also includes the epitaxial layer **508a** (e.g., a first epitaxial layer) extending into the fin structure **206a** and the epitaxial layer **508b** (e.g., a second epitaxial layer) extending into the fin structure **206b**. Furthermore, the semiconductor device **200** includes the insulator layer **304a** (e.g., a first insulator layer) between the fin structure **206a** and a bottom portion of the epitaxial layer **508a** and also includes the insulator layer **304b** between the fin structure **206b** and a bottom portion of the epitaxial layer **508b**.

(86) The number and arrangement of processing operations shown in FIGS. **5A-5F** are provided as one or more examples. In practice, there may be additional processing operations, fewer processing operations, different processing operations, or differently arranged processing operations than those shown in FIGS. **5A-5F**. For example, additional or different processing operations may include masking and/or exposing portions of the device region **202** using a combination of the deposition tool **102**, the exposure tool **104**, and the developer tool **106**. As another example, additional or different processing operations may include using the ion implantation tool **114** of FIG. **1** to implant ions and dope the epitaxial layers **508**, **510**, and/or **516**.

(87) FIGS. **6A-6C** are diagrams **600** of example types of devices described herein. The devices described in connection with diagrams **600** may be formed using one or more techniques described in connection with FIGS. **3A-3H**, **4A-4C**, and **5A-5F**. Furthermore, FIGS. **6A-6C** are illustrated from the perspective of the cross-sectional plane A-A in FIG. **2** for the device region **202**.

(88) FIG. **6A** shows an example **602** that corresponds to an SRAM type of semiconductor device (e.g., a type of the semiconductor device **200**). The example **602** includes a fin structure **206c** (e.g., the fin structure **206** extending above the semiconductor substrate **204**), an epitaxial layer **508c** (e.g., a first epitaxial layer) extending into the fin structure **206c**, and another epitaxial layer **508d** (e.g., a second epitaxial layer) extending into the fin structure **206c** adjacent to the epitaxial layer **508c**. The gate structure **212a** is between the epitaxial layer **508c** and the epitaxial layer **508d**.

(89) The SRAM type of semiconductor device includes an insulator layer **304c** (e.g., a first oxide layer or a first dielectric layer, among other examples) between a bottom surface **604a** (e.g., a first bottom surface) of the epitaxial layer **508c** and the fin structure **206c**. The SRAM type of semiconductor device further includes another insulator layer **304d** (e.g., a second oxide layer or a second dielectric layer, among other examples) between a bottom surface **604b** (e.g., a second bottom surface) of the epitaxial layer **508d** and the fin structure **206c**.

(90) At least one of the insulator layer **304c** or the insulator layer **304d** includes a thickness **606a** and a width **608a**. The thickness **606a** may be in a range from approximately 3 nanometers to approximately 13 nanometers, and the width **608a** may be in a range from approximately 10 nanometer to approximately 30 nanometers. For a thickness **606a** and a width **608a** that are less than these ranges, leakage within a transistor of the SRAM type of semiconductor device may increase. For a thickness **606a** and a width **608a** that are greater than these ranges, an activation voltage for the transistor of the SRAM type of semiconductor device may increase beyond operability. However, other values and ranges for the thickness **606a** and the width **608a** are within the scope of the present disclosure.

(91) FIG. **6B** shows an example **610** that corresponds to a RO type of semiconductor device (e.g., a type of the semiconductor device **200**). The example **610** includes a fin structure **206d** (e.g., the fin structure **206** extending above the semiconductor substrate **204**), an epitaxial layer **508e** (e.g., a first epitaxial layer) extending into the fin structure **206d**, and another epitaxial layer **508f** (e.g., a

second epitaxial layer) extending into the fin structure **206d** adjacent to the epitaxial layer **508e**.

The gate structure **212b** is between epitaxial layer **508e** and the epitaxial layer **508f**.

(92) The RO type of semiconductor device includes an insulator layer **304e** (e.g., a first oxide layer) between a bottom surface **604c** (e.g., a first bottom surface) of the epitaxial layer **508e** and the fin structure **206d**. The RO type of semiconductor device further includes another insulator layer **304f** (e.g., a second oxide layer) between a bottom surface **604d** (e.g., a second bottom surface) of the epitaxial layer **508f** and the fin structure **206d**.

(93) At least one of the insulator layer **304e** or the insulator layer **304f** includes a thickness **606b** and a width **608b**. The thickness **606b** may be in a range from approximately 1 nanometer to approximately 5 nanometers, and the width **608b** may be in a range from approximately 10 nanometers to approximately 40 nanometers. For a thickness **606b** and width **608b** that are less than these ranges, leakage within a transistor of the RO type of semiconductor device increase. For a thickness **606b** and a width **608b** that are greater than these ranges, an activation voltage for the transistor of the RO type of semiconductor device may increase beyond operability. However, other values and ranges for the thickness **606b** and the width **608b** are within the scope of the present disclosure.

(94) FIG. **6C** shows an example **612** that corresponds to an IO type of semiconductor device (e.g., a type of the semiconductor device **200**). The example **612** includes a fin structure **206e** (e.g., the fin structure **206** extending above the semiconductor substrate **204**), an epitaxial layer **508g** (e.g., a first epitaxial layer) extending into the fin structure **206e**, and another epitaxial layer **508h** (e.g., a second epitaxial layer) extending into the fin structure **206e** adjacent to the epitaxial layer **508g**. The gate structure **212c** is between epitaxial layer **508g** and the epitaxial layer **508h**.

(95) The IO type of semiconductor device includes an insulator layer **304g** (e.g., a first oxide layer or a first dielectric layer, among other examples) between a bottom surface **604e** (e.g., a first bottom surface) of the epitaxial layer **508g** and the fin structure **206e**. The IO type of semiconductor device further includes another insulator layer **304h** (e.g., a second oxide layer or a second dielectric layer, among other examples) between a bottom surface **604f** (e.g., a second bottom surface) of the epitaxial layer **508h** and the fin structure **206e**.

(96) At least one of the insulator layer **304g** or the insulator layer **304h** includes a thickness **606c** and a width **608c**. The thickness **606c** may be in a range from approximately 1 nanometer to approximately 7 nanometers, and the width **608c** may be in a range from approximately 20 nanometers to approximately 60 nanometers. For a thickness **606c** and width **608c** that are less than these ranges, leakage within a transistor of the RO type of semiconductor device may increase. For a thickness **606c** and a width **608c** that are greater than these ranges, an activation voltage for the transistor of the IO type of semiconductor device may increase beyond operability. However, other values and ranges for the thickness **606c** and the width **608c** are within the scope of the present disclosure.

(97) The number and arrangement of devices shown in FIG. **6A-6C** are provided as one or more examples. In practice, there may be additional devices, fewer devices, different devices, or differently arranged devices than those shown in FIGS. **6A-6C**. Furthermore, two or more devices shown in FIGS. **6A-6C** may be implemented within a single device, or a single device shown in FIGS. **6A-6C** may be implemented as multiple, distributed devices.

(98) FIGS. **7A-7C** are diagrams **700** of example processing operations to form the insulator layer **304** described herein (e.g., forming a dielectric layer including a layer of a silicon dioxide (SiO₂) material, among other examples). FIGS. **7A-7C** are illustrated from the perspective of the cross-sectional plane B-B in FIG. **2** for the device region **202**. The insulator layer **304** may be formed in or on the semiconductor substrate **204** prior to deposition of the epitaxial layer **508**. In some implementations, the structure produced by the techniques described in FIGS. **7A-7C** corresponds to a silicon-on-insulator (SOI) substrate.

(99) FIG. **7A** shows a method of forming the insulator layer **304**. As shown, an ion-implantation

processing operation **702** (e.g., an ion-implantation processing operation performed by the ion implantation tool **114**) may include implanting ions **704** into the semiconductor substrate **204** to form the insulator layer **304**. Implanting the ions may include implanting oxygen (O) ions, boron (B) ions, arsenic (As) ions, or phosphorous (P) ions, among other examples. In some implementations, one or more additional layers (e.g., a cover layer or a patterned masking layer, among other examples) may be over the semiconductor substrate **204** to create regions of the insulator layer **304** within the semiconductor substrate **204**.

(100) In some implementations, a thickness **706** of the insulator layer **304** may be in a range from approximately 5 nanometers to approximately 50 nanometers. For a thickness **706** that is greater than this range, subsequent processing operations for a selected type of the semiconductor device **200** (e.g., an SRAM type of semiconductor device, an RO type of semiconductor device, or an IO type of semiconductor device, among other examples) may not produce a transistor having a targeted activation voltage. For a thickness **706** that is less than this range, subsequent processing operations for the selected type of the semiconductor device **200** may not be able to produce a transistor with reduced leakage characteristics. However, other values and ranges for the thickness **706** are within the scope of the present disclosure.

(101) FIG. 7B shows another method of forming the insulator layer **304**. At processing operation **708** of FIG. 7B, the insulator layer **304** is formed on a portion of the semiconductor substrate **204a**. Forming the insulator layer **304** at processing operation **708** may include, as an example, forming the insulator layer **304** using a thermal-oxidation processing operation or a CVD processing operation.

(102) After formation of the insulator layer **304**, and at processing operation **710**, another portion of the semiconductor substrate **204b** is formed. Processing operation **710** may include forming the portion of the semiconductor substrate **204b** on the insulator layer **304** using an epitaxial-growth processing operation. The processing operation **710** may use a silicon containing gas such as a silane (SiH.sub.4) gas and/or a germanium containing gas such as germane (GeH.sub.4) gas, among other examples. Using such gases, the processing operation **710** (e.g., an epitaxial-growth processing operation) may epitaxially grow the portion of the semiconductor substrate **204b** (e.g., an epitaxial layer) from a layer of an Si material or a layer of an SiGe material, among other examples.

(103) FIG. 7C shows another method of forming the insulator layer **304**. At processing operation **712** (e.g., after forming the recesses **502** as described in connection with FIG. 5B), portions of the insulator layer **304** may be formed through oxidation growth. The oxidation growth may include, for example, native growth of SiO.sub.2 on bottom surfaces of the recesses **502**, deposition of SiO.sub.2 on bottom surfaces of the recesses **502**, and/or another oxidation growth technique.

(104) As described in FIGS. 7A-7C, and in connection with processing operations described in FIGS. 3A-3H, 4A-4D, and 5A-5F, a method that forms the semiconductor device **200** with the epitaxial region **512** over the insulator layer **304** may be performed. The method may include forming the insulator layer **304** (e.g., a dielectric layer) in or on the semiconductor substrate **204** and forming the fin structure **206** in the semiconductor substrate **204**. In some implementations, the fin structure **206** includes a portion of the insulator layer **304**. The method may further include forming the recess **502** in the fin structure **206** to the portion of the insulator layer **304** and forming the epitaxial region **512** over the portion of the insulator layer **304** in the recess **502**.

(105) As indicated above, FIGS. 7A-7C are provided as examples. Other examples may differ from what is described with regard to FIGS. 7A-7C. For example, different combinations of the processing operations described by FIGS. 7A-7C may be used to form a first insulator layer (e.g., the insulator layer **304a** of FIGS. 5A-5F, among other examples) from a first material and to form a second insulator layer (e.g., the insulator layer **304b** of FIGS. 5A-5F, among other examples) from a second material that is different than the first material. Additionally, or alternatively, different combinations of the processing operations described by FIGS. 7A-7C may be used to form a first

insulator layer (e.g., the insulator layer **304c** of FIG. **6A**, among other examples) to a first thickness (e.g., the thickness **606a**) and to form a second insulator layer (e.g., the insulator layer **304d** of FIG. **6A**, among other examples) to a second thickness (e.g., the thickness **606a**) that is different than the first thickness.

(106) FIG. **8** is a diagram of an example elemental analysis **800** of portions of the semiconductor device **200** described herein. FIG. **8** illustrates an example vertical elemental analysis **802** of the fin structure **206** through a source/drain region including the epitaxial layer **508** and the insulator layer **304**. The vertical elemental analysis **802**, which may be performed using an energy dispersive x-ray (EDX) analysis technique, is along a vertical axis **804**.

(107) As shown in FIG. **8**, a normalized absolute intensity of a signal **806** corresponding to an amount of silicon along a portion of the vertical axis **804** (e.g., a portion corresponding to the epitaxial layer **508**) is in a range from approximately 80% to approximately 100%. Along the same portion, the normalized absolute intensity of a signal **808** corresponding to an amount of oxygen is in a range from approximately 20% to approximately 30%.

(108) As further shown in FIG. **8**, the normalized absolute intensity of the signal **806** reduces to a range from approximately 60% to 70% along another portion of the vertical axis **804** (e.g., another portion corresponding to the insulator layer **304**). The reduction of the normalized absolute intensity of the signal **806** in this other portion corresponds to a reduced presence of silicon.

(109) Along the same other portion, the normalized absolute intensity of the signal **808** increases to a range from approximately 35% to approximately 45%. The increase in the normalized absolute intensity of the signal **808** in this other portion corresponds to an enriched presence of oxygen (e.g., oxygen included in SiO₂, among other examples).

(110) As indicated above, FIG. **8** is provided as an example. Other examples may differ from what is described with regard to FIG. **8**.

(111) FIG. **9** is a diagram of example components of a device **900** described herein. In some implementations, one or more of the semiconductor processing devices **102-114** and/or the wafer/die transport tool **116** may include one or more devices **900** and/or one or more components of device **900**.

(112) As shown in FIG. **9**, device **900** may include a bus **910**, a processor **920**, a memory **930**, an input component **940**, an output component **950**, and a communication component **960**.

(113) Bus **910** includes one or more components that enable wired and/or wireless communication among the components of device **900**. Bus **910** may couple together two or more components of FIG. **9**, such as via operative coupling, communicative coupling, electronic coupling, and/or electric coupling. Processor **920** includes a central processing unit, a graphics processing unit, a microprocessor, a controller, a microcontroller, a digital signal processor, a field-programmable gate array, an application-specific integrated circuit, and/or another type of processing component. Processor **920** is implemented in hardware, firmware, or a combination of hardware and software. In some implementations, processor **920** includes one or more processors capable of being programmed to perform one or more operations or processes described elsewhere herein.

(114) Memory **930** includes volatile and/or nonvolatile memory. For example, memory **930** may include random access memory (RAM), read only memory (ROM), a hard disk drive, and/or another type of memory (e.g., a flash memory, a magnetic memory, and/or an optical memory). Memory **930** may include internal memory (e.g., RAM, ROM, or a hard disk drive) and/or removable memory (e.g., removable via a universal serial bus connection). Memory **930** may be a non-transitory computer-readable medium. Memory **930** stores information, instructions, and/or software (e.g., one or more software applications) related to the operation of device **900**. In some implementations, memory **930** includes one or more memories that are coupled to one or more processors (e.g., processor **920**), such as via bus **910**.

(115) Input component **940** enables device **900** to receive input, such as user input and/or sensed input. For example, input component **940** may include a touch screen, a keyboard, a keypad, a

mouse, a button, a microphone, a switch, a sensor, a global positioning system sensor, an accelerometer, a gyroscope, and/or an actuator. Output component **950** enables device **900** to provide output, such as via a display, a speaker, and/or a light-emitting diode. Communication component **960** enables device **900** to communicate with other devices via a wired connection and/or a wireless connection. For example, communication component **960** may include a receiver, a transmitter, a transceiver, a modem, a network interface card, and/or an antenna.

(116) Device **900** may perform one or more operations or processes described herein. For example, a non-transitory computer-readable medium (e.g., memory **930**) may store a set of instructions (e.g., one or more instructions or code) for execution by processor **920**. Processor **920** may execute the set of instructions to perform one or more operations or processes described herein. In some implementations, execution of the set of instructions, by one or more processors **920**, causes the one or more processors **920** and/or the device **900** to perform one or more operations or processes described herein. In some implementations, hardwired circuitry is used instead of or in combination with the instructions to perform one or more operations or processes described herein. Additionally, or alternatively, processor **920** may be configured to perform one or more operations or processes described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.

(117) The number and arrangement of components shown in FIG. **9** are provided as an example. Device **900** may include additional components, fewer components, different components, or differently arranged components than those shown in FIG. **9**. Additionally, or alternatively, a set of components (e.g., one or more components) of device **900** may perform one or more functions described as being performed by another set of components of device **900**.

(118) FIG. **10** is a flowchart of an example process **1000** associated with forming a semiconductor device described herein. In some implementations, one or more process blocks of FIG. **10** are performed by one or more semiconductor processing tools (e.g., one or more of the semiconductor processing tools **102-114**). Additionally, or alternatively, one or more process blocks of FIG. **10** may be performed by one or more components of device **900**, such as processor **920**, memory **930**, input component **940**, output component **950**, and/or communication component **960**.

(119) As shown in FIG. **10**, process **1000** may include forming a dielectric layer in or on a semiconductor substrate (block **1010**). For example, one or more of the processing tools **102-114** may form a dielectric layer (e.g., an insulator layer **304**) in or on a semiconductor substrate **204**, as described above.

(120) As further shown in FIG. **10**, process **1000** may include forming a fin structure in the semiconductor substrate (block **1020**). For example, one or more of the semiconductor processing tools **102-114** may form a fin structure **206** in the semiconductor substrate **204**. In some implementations, the fin structure **206** includes a portion of the dielectric layer.

(121) As further shown in FIG. **10**, process **1000** may include forming a recess in the fin structure to the portion of the dielectric layer (block **1030**). For example, one or more of the semiconductor processing tools **102-114** may form a recess **502** in the fin structure **206** to the portion of the dielectric layer, as described above.

(122) As further shown in FIG. **10**, process **1000** may include forming an epitaxial region over the portion of the dielectric layer in the recess (block **1040**). For example, one or more of the semiconductor processing tools **102-114** may form an epitaxial region **512** over the portion of the dielectric layer in the recess **502**, as described above.

(123) Process **1000** may include additional implementations, such as any single implementation or any combination of implementations described below and/or in connection with one or more other processes described elsewhere herein.

(124) In a first implementation, forming the dielectric layer in or on the semiconductor substrate **204** includes forming the dielectric layer to a thickness **706** that is in a range from approximately 5 nanometers to approximately 50 nanometers.

(125) In a second implementation, alone or in combination with the first implementation, forming the dielectric layer in or on the semiconductor substrate **204** includes forming a layer of silicon dioxide (SiO₂) material.

(126) In a third implementation, alone or in combination with one or more of the first and second implementations, forming the dielectric layer in or on the semiconductor substrate **204** includes forming the dielectric layer below a surface of the semiconductor substrate **204** using an ion-implantation processing operation **702**.

(127) In a fourth implementation, alone or in combination with one or more of the first through third implementations, forming the recess **502** in the fin structure **206** includes forming the recess **502** into a portion of the dielectric layer.

(128) In a fifth implementation, alone or in combination with one or more of the first through fourth implementations, forming the dielectric layer in or on the semiconductor substrate **204** includes forming the dielectric layer on a surface of the semiconductor substrate **204** using a thermal-oxidation processing operation **708**.

(129) In a sixth implementation, alone or in combination with one or more of the first through fifth implementations, process **1000** includes forming an epitaxial layer (e.g., a semiconductor substrate **204b**) on a top surface of the dielectric layer using an epitaxial-growth processing operation **710** prior to forming the fin structure **206**.

(130) In a seventh implementation, alone or in combination with one or more of the first through sixth implementations, forming the fin structure **206** includes forming the fin structure **206** such that the fin structure **206** includes a portion of the epitaxial layer (e.g., a semiconductor substrate **204b**).

(131) In an eighth implementation, alone or in combination with one or more of the first through seventh implementations, forming the epitaxial region **512** includes forming a first epitaxial layer **508** on the dielectric layer and between fin sidewall spacers **504**, forming a second epitaxial layer **510** on the first epitaxial layer **508** and above the fin sidewall spacers **504**, and forming a third epitaxial layer **516** on the second epitaxial layer **510**.

(132) In a ninth implementation, alone or in combination with one or more of the first through eighth implementations, forming the second epitaxial layer **510** includes forming a merged-epitaxial sub-region **514** that joins the epitaxial region and another epitaxial region **512b** on another fin structure **206b**.

(133) In a tenth implementation, alone or in combination with one or more of the first through ninth implementations, process **1000** includes forming a contact layer **520** over the third epitaxial layer **516**.

(134) Although FIG. **10** shows example blocks of process **1000**, in some implementations, process **1000** includes additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **10**. Additionally, or alternatively, two or more of the blocks of process **1000** may be performed in parallel.

(135) Some implementations described herein provide techniques and apparatuses for forming insulator layers in or on a semiconductor substrate prior to forming epitaxial layers within source/drain regions of a finFET. The epitaxial layers may be formed over the insulator layers to reduce electron tunneling between the source/drain regions of the finFET.

(136) In this way, a likelihood of leakage into the semiconductor substrate and/or between the source/drain regions of the finFET is reduced. As a result, performance characteristics relating to power consumption of a semiconductor device including the finFET and operation of a gate structure between the source/drain regions may be increased.

(137) As described in greater detail above, some implementations described herein provide a semiconductor device. The semiconductor device includes a fin structure above a semiconductor substrate. The semiconductor device includes a first epitaxial region extending into the fin structure. The semiconductor device includes a second epitaxial region extending into the fin

structure adjacent to the first epitaxial region. The semiconductor device includes a gate structure between the first epitaxial region and the second epitaxial region. The semiconductor device includes a first oxide layer between a first bottom surface of the first epitaxial region and the fin structure. The semiconductor device includes a second oxide layer between a second bottom surface of the second epitaxial region and the fin structure.

(138) As described in greater detail above, some implementations described herein provide a method. The method includes forming a dielectric layer in or on a semiconductor substrate. The method includes forming a fin structure in the semiconductor substrate, where the fin structure includes a portion of the dielectric layer. The method includes forming a recess in the fin structure to the portion of the dielectric layer. The method includes forming an epitaxial region over the portion of the dielectric layer in the recess.

(139) As described in greater detail above, some implementations described herein provide a semiconductor device. The semiconductor device includes a first fin structure above a semiconductor substrate. The semiconductor device includes a second fin structure above the semiconductor substrate and adjacent to the first fin structure. The semiconductor device includes a first epitaxial region extending into the first fin structure. The semiconductor device includes a second epitaxial region extending into the second fin structure. The semiconductor device includes a first insulator layer between the first fin structure and a bottom portion of the first epitaxial region. The semiconductor device includes a second insulator layer between the second fin structure and a bottom portion of the second epitaxial region.

(140) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method, comprising: forming a dielectric layer in or on a semiconductor substrate; forming a fin structure in the semiconductor substrate, wherein the fin structure includes a portion of the dielectric layer; forming a recess in the fin structure to an entire top surface of the portion of the dielectric layer, wherein a bottom of the recess exposes the entire top surface of the portion of the dielectric layer; and forming an epitaxial region over the entire top surface of the portion of the dielectric layer in the recess.
2. The method of claim 1, wherein forming the dielectric layer in or on the semiconductor substrate comprises: forming the dielectric layer to a thickness that is in a range from approximately 5 nanometers to approximately 50 nanometers.
3. The method of claim 1, wherein forming the dielectric layer in or on the semiconductor substrate comprises: forming a layer of silicon dioxide (SiO₂) material.
4. The method of claim 1, wherein forming the dielectric layer in or on the semiconductor substrate comprises: forming the dielectric layer below a surface of the semiconductor substrate using an ion-implantation processing operation.
5. The method of claim 1, wherein forming the dielectric layer in or on the semiconductor substrate comprises: forming the dielectric layer on a surface of the semiconductor substrate using a thermal-oxidation processing operation.
6. The method of claim 1, further comprising: forming an epitaxial layer on a top surface of the dielectric layer using an epitaxial-growth processing operation prior to forming the fin structure.

7. The method of claim 6, wherein forming the fin structure comprises: forming the fin structure such that the fin structure includes a portion of the epitaxial layer.
 8. The method of claim 1, wherein forming the epitaxial region comprises: forming a first epitaxial layer on the dielectric layer and between fin sidewall spacers; forming a second epitaxial layer on the first epitaxial layer and above the fin sidewall spacers; and forming a third epitaxial layer on the second epitaxial layer.
 9. The method of claim 8, wherein forming the second epitaxial layer comprises: forming a merged-epitaxial sub-region that joins the epitaxial region and another epitaxial region on another fin structure.
 10. The method of claim 8, further comprising: forming a contact layer over the third epitaxial layer.
 11. A method, comprising: forming one or more fin structures extending from a substrate and between hybrid fin structures, wherein the one or more fin structures reside above an insulator layer that resides in the substrate; forming one or more recesses in the one or more fin structures to one or more portions of the insulator layer; and forming one or more epitaxial regions in the one or more recesses and over an entire top surface of the one or more portions of the insulator layer.
 12. The method of claim 11, further comprising: forming sidewall spacers on the one or more fin structures, wherein the one or more recesses are further formed in the sidewall spacers.
 13. The method of claim 12, further comprising: forming one or more epitaxial layers within the one or more fin structures, wherein a top surface of the one or more epitaxial layers is below a top surface of the one or more fin structures, and wherein the one or more epitaxial regions are formed over the one or more epitaxial layers.
 14. The method of claim 11, wherein after the one or more recesses are formed, a top surface of the one or more recesses is rounded or concave.
 15. The method of claim 11, wherein forming the one or more epitaxial regions comprises: forming a first epitaxial layer on the insulator layer, and forming a second epitaxial layer on the first epitaxial layer.
 16. The method of claim 15, wherein: the first epitaxial layer is formed within side spacers of the one or more fin structures, and the second epitaxial layer is formed above the side spacers.
 17. The method of claim 15, wherein the second epitaxial layer is more heavily doped than the first epitaxial layer.
 18. A method, comprising: forming a plurality of fin structures extending from a substrate and between a plurality of hybrid fin structures, wherein the plurality of fin structures resides above an insulator layer that resides in the substrate; forming a first recess, in a first fin structure of the plurality of fin structures to an entire top surface of a first portion of the insulator layer, and a second recess, in a second fin structure of the plurality of fin structures to an entire top surface of a second portion of the insulator layer, wherein a bottom of the first recess exposes the entire top surface of the first portion of the insulator layer, and wherein a bottom of the second recess exposes the entire top surface of the second portion of the insulator layer; and forming a merged-epitaxial sub-region that joins a first epitaxial region, over the first recess, and a second epitaxial region, over the second recess.
 19. The method of claim 18, further comprising: forming an epitaxial layer over the first epitaxial region, the second epitaxial region, and the merged-epitaxial sub-region.
 20. The method of claim 18, further comprising at least one of: forming an inner-layer dielectric (ILD) layer over the plurality of hybrid fin structures, or forming a contact layer, comprising titanium, over a top surface of the first epitaxial region, the second epitaxial region, and the merged-epitaxial sub-region.
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